Banking Application

Java Lab – Assignment 5

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GITHUB: https://github.com/kavanpatel18/Programming-with-Java

Customer.java

```
public class Customer {
    private String customerID;
    private String firstName;
    private String lastName;
    private String email;
    private String phoneNumber;
    private String address;
    protected String aadharID;
    protected String panID;

    public Customer(String customerID, String firstName, String lastName,
                    String email, String phoneNumber, String address,
                    String aadharID, String panID) {
        this.customerID  = customerID;
        this.firstName   = firstName;
        this.lastName    = lastName;
        this.email       = email;
        this.phoneNumber = phoneNumber;
        this.address     = address;
        this.aadharID    = aadharID;
        this.panID       = panID;
    }

    public String getCustomerID() { return customerID; }

    public void displayCustomer() {
        System.out.println("Customer ID : " + customerID);
        System.out.println("Name        : " + firstName + " " + lastName);
        System.out.println("Email       : " + email);
        System.out.println("Phone       : " + phoneNumber);
        System.out.println("Address     : " + address);
        System.out.println("Aadhar ID   : " + aadharID);
        System.out.println("PAN ID      : " + panID);
    }
}
```

InsufficientBalanceException.java

```
public class InsufficientBalanceException extends Exception {
    public InsufficientBalanceException(String message) {
        super(message);
    }
}
```

InvalidAccountException.java

```
public class InvalidAccountException extends Exception {
    public InvalidAccountException(String message) {
        super(message);
    }
}
```

Account.java

```
public class Account {
    protected int     accountNumber;
    protected double  balance;
    protected Customer customer;

    public Account(int accountNumber, double balance, Customer customer) {
        this.accountNumber = accountNumber;
        this.balance       = balance;
        this.customer      = customer;
    }

    public void deposit(double amount) {
        if (amount <= 0) {
            System.out.println("Deposit amount must be positive.");
            return;
        }
        balance += amount;
        System.out.printf("Deposited : Rs. %.2f  |  New Balance : Rs. %.2f%n", amount,
balance);
    }

    public void withdraw(double amount) throws InsufficientBalanceException {
        if (amount > balance)
            throw new InsufficientBalanceException("Insufficient balance.");
        balance -= amount;
        System.out.printf("Withdrawn : Rs. %.2f  |  Remaining Balance : Rs. %.2f%n",
amount, balance);
    }

    public void displayAccount() {
        customer.displayCustomer();
        System.out.println("Account Number : " + accountNumber);
        System.out.printf("Balance        : Rs. %.2f%n", balance);
    }

    public double   getBalance()       { return balance; }
    public int      getAccountNumber() { return accountNumber; }
    public Customer getCustomer()      { return customer; }
}
```

SavingsAccount.java

```
public class SavingsAccount extends Account {
    private static final double INTEREST_RATE   = 0.067;   // 6.7% p.a.
    private static final double MINIMUM_BALANCE = 500.0;
```

```
    public SavingsAccount(int accountNumber, double balance, Customer customer) {
        super(accountNumber, balance, customer);
    }

    public void withdraw(double amount) throws InsufficientBalanceException {
        if (balance - amount < MINIMUM_BALANCE)
            throw new InsufficientBalanceException(
                "Withdrawal denied: minimum balance of Rs. "
                + MINIMUM_BALANCE + " must be maintained.");
        balance -= amount;
        System.out.printf("[Savings] Withdrawn : Rs. %.2f  |  Balance : Rs. %.2f%n",
                          amount, balance);
    }

    public void calculateInterest() {
        double interest = balance * INTEREST_RATE;
        System.out.printf("Interest Earned (%.1f%%) : Rs. %.2f%n",
                          INTEREST_RATE * 100, interest);
    }
}
```

LoanAccount.java

```
public class LoanAccount extends Account {
    private double loanAmount;
    private int    totalMonths;
    public  double monthlyEMI;
    private int    monthsPaid;

    public LoanAccount(int accountNumber, double loanAmount,
                       int totalMonths, Customer customer) {
        super(accountNumber, 0, customer);
        this.loanAmount  = loanAmount;
        this.totalMonths = totalMonths;
        this.monthlyEMI  = loanAmount / totalMonths;
        this.monthsPaid  = 0;
    }


    public void deposit(double amount) {
        double outstanding = loanAmount - (monthsPaid * monthlyEMI);
        if (outstanding <= 0) {
            System.out.println("Loan already fully repaid.");
            return;
        }
        monthsPaid++;
        outstanding -= amount;
        System.out.printf("EMI Paid : Rs. %.2f  |  Outstanding Loan : Rs. %.2f%n",
                          amount, Math.max(0, outstanding));
    }

    public void withdraw(double amount) throws InsufficientBalanceException {
        throw new InsufficientBalanceException(
            "Withdrawals are not allowed on Loan Accounts.");
    }

    public void displayLoan() {
        System.out.println("--- Loan Account Details ---");
        customer.displayCustomer();
        System.out.println("Account Number : " + accountNumber);
```

```
        System.out.printf("Loan Amount    : Rs. %.2f%n", loanAmount);
        System.out.printf("Monthly EMI    : Rs. %.2f%n", monthlyEMI);
        System.out.printf("Tenure         : %d months%n", totalMonths);
        System.out.printf("Months Paid    : %d%n", monthsPaid);
        System.out.printf("Outstanding    : Rs. %.2f%n",
                          Math.max(0, loanAmount - monthsPaid * monthlyEMI));
    }
}
```

Main.java

```
import java.util.ArrayList;

public class Main {

    public static void main(String[] args) {
        try {
            // 1. Create Customers
            ArrayList<Customer> customers = new ArrayList<>();

            Customer c1 = new Customer("C001", "Kavan", "Patel",
                "kavan.patel@gmail.com", "9829326907", "Pune",
                "1234 5678 9101", "ABCDE1234F");

            Customer c2 = new Customer("C002", "Arjun", "Mehta",
                "arjun.mehta@gmail.com", "9876543210", "Mumbai",
                "9876 5432 1098", "XYZAB5678G");

            Customer c3 = new Customer("C003", "Sneha", "Rao",
                "sneha.rao@gmail.com", "9123456789", "Delhi",
                "1234 1234 1234", "PQRST9012H");

            customers.add(c1);
            customers.add(c2);
            customers.add(c3);

            // 2. Create Accounts
            ArrayList<Account> accounts = new ArrayList<>();

            SavingsAccount sa1 = new SavingsAccount(1001, 10000, c1);
            LoanAccount    la1 = new LoanAccount(3001, 120000, 12, c1);
            accounts.add(sa1);
            accounts.add(la1);

            SavingsAccount sa2 = new SavingsAccount(1002, 25000, c2);
            LoanAccount    la2 = new LoanAccount(3002, 60000, 6, c2);
            accounts.add(sa2);
            accounts.add(la2);

            SavingsAccount sa3 = new SavingsAccount(1003, 5000, c3);
            accounts.add(sa3);

            // 3. Perform Transactions
            System.out.println("\n========= Performing Transactions =========");

            System.out.println("\n--- Savings Account (Kavan) ---");
            sa1.deposit(5000);
            sa1.withdraw(3000);
            sa1.calculateInterest();
```

```
            System.out.println("\n--- Loan Account (Kavan) ---");
            la1.displayLoan();
            la1.deposit(la1.monthlyEMI);

            System.out.println("\n--- Savings Account (Arjun) ---");
            sa2.deposit(10000);
            sa2.withdraw(5000);
            sa2.calculateInterest();

            // 4. Consolidated Summary
            System.out.println("\n========= Customer & Account Summary =========");
            displayConsolidatedInfo(customers, accounts);

        } catch (InsufficientBalanceException e) {
            System.out.println("Exception: " + e.getMessage());
        }
    }

    public static void displayConsolidatedInfo(ArrayList<Customer> customers,
                                               ArrayList<Account>  accounts) {
        for (Customer cust : customers) {
            System.out.println("\n==== CUSTOMER DETAILS ====");
            cust.displayCustomer();
            System.out.println("\n---- ACCOUNTS ----");

            double totalBalance = 0;
            int    accountCount = 0;

            for (Account acc : accounts) {
                if (acc.getCustomer().getCustomerID().equals(cust.getCustomerID())) {
                    System.out.println("Account Number : " + acc.getAccountNumber());
                    System.out.println("Account Type   : " +
acc.getClass().getSimpleName());
                    System.out.printf("Balance        : Rs. %.2f%n", acc.getBalance());
                    System.out.println();
                    totalBalance += acc.getBalance();
                    accountCount++;
                }
            }
            System.out.println("Total Accounts                    : " + accountCount);
            System.out.printf("Total Balance Across All Accounts : Rs. %.2f%n",
totalBalance);
            System.out.println("--------------------------------------------------");
        }
    }
}
```

OUTPUT:

```
========= Performing Transactions =========

--- Savings Account (Kavan) ---
Deposited : Rs. 5000.00  |  New Balance : Rs. 15000.00
[Savings] Withdrawn : Rs. 3000.00  |  Balance : Rs. 12000.00
Interest Earned (6.7%) : Rs. 804.00

--- Loan Account (Kavan) ---
--- Loan Account Details ---
Customer ID : C001
Name        : Kavan Patel
Email       : kavan.patel@gmail.com
Phone       : 9829326907
Address     : Pune
Aadhar ID   : 1234 5678 9101
PAN ID      : ABCDE1234F
Account Number : 3001
Loan Amount    : Rs. 120000.00
Monthly EMI    : Rs. 10000.00
Tenure         : 12 months
Months Paid    : 0
Outstanding    : Rs. 120000.00
EMI Paid : Rs. 10000.00  |  Outstanding Loan : Rs. 110000.00

--- Savings Account (Arjun) ---
Deposited : Rs. 10000.00  |  New Balance : Rs. 35000.00
[Savings] Withdrawn : Rs. 5000.00  |  Balance : Rs. 30000.00
Interest Earned (6.7%) : Rs. 2010.00
```

```
========= Customer & Account Summary =========

==== CUSTOMER DETAILS ====
Customer ID : C001
Name        : Kavan Patel
Email       : kavan.patel@gmail.com
Phone       : 9829326907
Address     : Pune
Aadhar ID   : 1234 5678 9101
PAN ID      : ABCDE1234F

---- ACCOUNTS ----
Account Number : 1001
Account Type   : SavingsAccount
Balance        : Rs. 12000.00

Account Number : 3001
Account Type   : LoanAccount
Balance        : Rs. 0.00

Total Accounts                    : 2
Total Balance Across All Accounts : Rs. 12000.00
--------------------------------------------------
```

```
==== CUSTOMER DETAILS ====
Customer ID : C002
Name        : Arjun Mehta
Email       : arjun.mehta@gmail.com
Phone       : 9876543210
Address     : Mumbai
Aadhar ID   : 9876 5432 1098
PAN ID      : XYZAB5678G

---- ACCOUNTS ----
Account Number : 1002
Account Type   : SavingsAccount
Balance        : Rs. 30000.00

Account Number : 3002
Account Type   : LoanAccount
Balance        : Rs. 0.00

Total Accounts                    : 2
Total Balance Across All Accounts : Rs. 30000.00
-------------------------------------------------

==== CUSTOMER DETAILS ====
Customer ID : C003
Name        : Sneha Rao
Email       : sneha.rao@gmail.com
Phone       : 9123456789
Address     : Delhi
Aadhar ID   : 1234 1234 1234
PAN ID      : PQRST9012H

---- ACCOUNTS ----
Account Number : 1003
Account Type   : SavingsAccount
Balance        : Rs. 5000.00

Total Accounts                    : 1
Total Balance Across All Accounts : Rs. 5000.00
-------------------------------------------------
```